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PI Name:	Marr, David W.M. Ph.D.		
Project Title:	Fabrication, Crystallization, and Folding of Complex Colloidal Molecules under the Influence of Applied External Fields		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80401-1887	**Congressional District:**	7
Comments:	https://		
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Flight Program:			
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COI Name (Institution):	Wu, Ning (Colorado School of Mines) Solomon, Michael (University of Michigan, Ann Arbor)		
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	Colloidal particles are technologically important materials with applications ranging from consumer products to plasmonic metamaterials to drug delivery agents. With phase behaviors and assembly characteristics that closely resemble atoms and molecules, they are also excellent molecular analogues that can grow into functional materials with high efficiency. The range of colloidal structures and molecular mimics we can obtain however is currently limited and we remain far from truly accessing the diversity of phases desired for both deeper fundamental understanding and associated technological application. This is due primarily to limited control of the intrinsic colloidal interactions in conventionally isotropic building blocks and a reliance on non-directed assembly. To address these issues, we propose the use of anisotropic external fields including optical and hydrodynamic fields to create anisotropic colloidal building blocks (primarily on ground). We also propose the use of anisotropic electric fields		

Task Description:	to direct their assembly into complex structures (primarily in space). This combination of rational design of colloidal composition and architecture and versatile control of the range, strength, and directionality of interparticle interactions will allow the study of two outstanding scientific questions particularly appropriate for investigation at the International Space Station: (i) How does complex symmetry evolve during the crystallization of anisotropic colloidal molecules, such as achiral dimer particles, racemic mixtures of chiral tetramers, and pure enantiomeric tetramers? (ii) How will the collective fluctuations of nodes (tetramers) and linkers (colloidal chains) induce the formation of mixed networks with non-linear mechanical properties? Results from these investigations will lead to better control of the non-covalent assembly of molecules and scalable processing of advanced materials with tailored structures and properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In-situ manipulation of anisotropic interactions and dynamic pathways, based on rational colloidal particle design and proper use of external fields, could lead to crystalline and aperiodic structures beyond those seen in nature. With the wide variety of attributes that characterize anisotropic interactions, we will probe previously inaccessible regions of complex phase space. Our studies will move significantly beyond hard spheres and towards experimental models appropriate for studying fundamental questions associated with complex symmetries. For example, how does chirality of the assembled structure evolve during the nucleation and crystallization of achiral molecules, chiral molecules in racemic mixtures, or in pure enantiomers? Furthermore, our models of giant colloidal chains will allow for the manipulation and observation of the effect of site-specific interactions on chain dynamics at experimentally accessible length and time scales. Models that can be easily visualized like these will eventually lead to better understanding of complex phenomena such as polymer chain dynamics, chain reptation, interpenetrating double network dynamics, and protein folding. Results from this study will lead to improved control of non-covalent assembly of molecules, efficient tailoring of lattice symmetries, and the scalable processing of nano-structured materials. The development of new colloidal molecules and their associated assembled structures may have considerable technological impact as well. Anisotropic interactions developed in this proposal, both at the particle level and via applied fields, have been shown generally to lead to arrays with reduced symmetry and enhanced directionality. Such structures interact with a broad range of electromagnetic radiation in unique ways and can exhibit collective photonic, plasmonic, mechanical, electronic, or magnetic properties that are not manifested at the level of single particles. As a result, they have significant potential as next-generation functional materials. For example, dielectric arrays with diamond-like or quasicrystalline lattices have been proposed as ideal photonic crystals that can manipulate light propagation in three dimensions, forming the basis for next-generation all-optical communication technologies. Chiral tetramers studied here could exhibit distinct plasmonic properties resulting from the collective coupling between particles in the pyramid, as confirmed in numerical modeling, which have great potential for electromagnetic metamaterials and high-resolution sensors. The understanding of mirror-symmetry breaking in racemates is also important for developing improved strategies for separation of pharmaceutical molecules. In addition, anisotropic particles can be used to generate new kinds of colloidal networks with well-defined coordination and connectivity. By combining families of tetramers and chains, double networks – two distinct, but interpenetrating, networks of nodes and linkers – could be formed with combined high mechanical stiffness and high toughness. Colloidal networks are very common in a range of pharmaceutical and consumer production applications because of their ability to impart a yield stress that can stabilize active components uniformly throughout the formulation. Developing methods to produce colloidal networks with defined coordination, connectivity, and topology would expand the use and performance of colloidal stabilizing networks in these materials. Ultimately, successful implementation of our proposed research will provide the fundamental knowledge necessary for the development of technologies to design and control matter with tailored structures and properties.
Task Progress:	1) We have successfully modified superparamagnetic spheres for the attachment of thiol functionality and employed a strong base as a catalyst to greatly increase the reaction rate of the thiol-epoxy "click" reaction. Change of surface charge from positive to negative was used to characterize extent of reaction and thiolate presence on the magnetic particle surface. 2) After thiol functional group surface attachment, we assembled colloids into linear chains by applying a DC magnetic field in the presence of linkers (bis PEG-maleimide or 4-arm PEG-maleimide). The Michael addition between thiols on the particle surface and maleimide on the linker molecules in solution creates chemical bonds between neighboring particles in ~30 min. 3) We have employed an analysis of chain assembly based on a classical step-growth polymerization where neighboring particles are regarded as monomers. After comparing chain distribution statistics from our experimental results with predictions from classical polymer theory, the bonding probability for neighboring particles was obtained under different reaction conditions. Results show that, under a fixed magnetic field, bonding probabilities vary slightly for different linker lengths. Similar probabilities for different reaction times show that the reaction is relatively fast with a strong temperature dependence over the 50°C vs. 75°C range investigated. 4) Chain stability was studied using polymer degradation theory. By measuring chain length distribution immediately after synthesis, experimental data was used to obtain decay coefficients. Here we find that the dependence of decay constants on reaction temperature, time, and linker length shows the same tendency as bonding probability, a significantly stronger dependence on reaction temperature than with reaction time and linker length. 5) We further characterized chain rigidity constants by balancing dipolar interactions between magnetic particles and elastic energies from the bending of magnetic chains. Our results showed that the longer the linker, the more flexible the magnetic chain and that, by changing linker length, chain flexibility can be tuned over two orders of magnitude. 6) In addition, we have observed that rotating magnetic fields can be employed to turn these magnetic chains into microlassos that can be used to isolate and transport cargo in microenvironments.
Bibliography Type:	Description: (Last Updated: 06/13/2025)

Abstracts for Journals and Proceedings	Yang T, Wu N, Marr DW. "Combined Magneto- and Optofluidic Assembly of Colloidal Chains of Controllable Length" Oral Presentation. 2015 AIChE National Meeting, Salt Lake City, November 8-13, 2015. 2015 AIChE National Meeting, Salt Lake City, November 8-13, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Colón-Meléndez L, Beltran-Villegas DJ, Yang T, Larson RG, Somomon MJ, Wu N, Marr D. "Kinetics of Lock-and-Key Binding for Fabrication of Complex Colloidal Molecules." Oral Presentation. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Yang T, Tasci O, Neeves K, Wu N, Marr DWM. "Microlasso Assembly and Control for Cargo Transport." Poster presentation. 2016 Smart and Active Matter Conference, Syracuse NY, June 20-23, 2016. 2016 Smart and Active Matter Conference, Syracuse NY, June 20-23, 2016. , Jun-2016